

Rajiv Gandhi University of Health Sciences, Karnataka

Fourth Semester B. Pharm Degree Examination – 16-Nov-2021

Time: Three Hours

Max. Marks: 75 Marks

MEDICINAL CHEMISTRY - I

Q.P. CODE: 5014

Your answers should be specific to the questions asked.

Draw neat labeled diagrams wherever necessary.

All the Questions are compulsory.

LONG ESSAYS

2 x 10 = 20 Marks

1. Define and classify antipsychotic agents with example. Write the SAR of phenothiazines and write the synthesis of Chlorpromazine.

OR

What are narcotic analgesics? Discuss in details the peripheral modifications in Morphine. Outline the synthesis of methadone.

2. a) What are adrenergic neurotransmitters? Give examples. Write their biosynthesis and metabolism.
b) Discuss the SAR of sympathomimetic agents.

SHORT ESSAYS

7 x 5 = 35 Marks

3. Explain the role of hydrogen bonding and partition coefficient on drug action.

OR

Explain in detail about factors effecting drug metabolism.

4. Explain the role of bio isosterism and geometrical isomerism on drug action.

OR

Write about direct acting cholinergic agent with examples. Write the synthesis of carbachol.

5. Write the synthesis of Diocyclomine HCl and Procyclidine HCl, give their uses.
6. Write the structure in uses of the following
 - a) Neostigmine
 - b) Ammonium chloride
 - c) Tropicamide
 - d) Pralidoxine chloride
 - e) Scopolamine
7. Define and classify general anesthetics with examples. Explain their mode of action.
8. Write a note on hydantoins used as anti convulsants. Outline the synthesis of any one.
9. Explain direct acting sympathomimetic agent with structures. Mention their uses. Write the synthesis of any one.

SHORT ANSWERS

10 x 2 = 20 Marks

10. Write the structure and uses of Ephedrine and Methyldopa.
11. Write the structure and uses of Ibuprofen and Diclofenac.
12. Give examples for cholinesterase inhibitors. Write the structure of any one.
13. Write the structure and uses of Parathion and Malathion.
14. Write the structure and functions of acetyl choline.
15. Give two examples for hydrolytic reactions in drug metabolism.
16. Write the importance of glucuronic acid in drug metabolism.
17. Write the synthesis and uses of Tolazoline.
18. What are mixed acting sympathomimetics? Give two structural examples.
19. Write the structure and uses of Metapralol and Prazocine.
